**EXTENSION 2**
**Complex Numbers (Ext2), N1 Introduction to Complex Numbers (Ext2)**

# Arithmetic of Complex Numbers

**Teacher:** Nardene Dodd
**Exam Equivalent Time:** 85.5 minutes *(based on allocation of 1.5 minutes per mark)*

## Questions

1. Complex Numbers, EXT2 N1 2012 HSC 1 MC

Let $z = 5 - i$ and $w = 2 + 3i$.

What is the value of $2z + \overline{w}$?

**(A)** $12 + i$

**(B)** $12 + 2i$

**(C)** $12 - 4i$

**(D)** $12 - 5i$

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2. Complex Numbers, EXT2 N1 2015 HSC 2 MC

What value of $z$ satisfies $z^2 = 7 - 24i$?

**A.** $4 - 3i$

**B.** $-4 - 3i$

**C.** $3 - 4i$

**D.** $-3 - 4i$

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3. Complex Numbers, EXT2 N1 2019 HSC 1 MC

What is the value of $(3 - 2i)^2$?

**A.** $5 - 12i$

**B.** $5 + 12i$

**C.** $13 - 12i$

**D.** $13 + 12i$

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4. Complex Numbers, EXT2 N1 2023 HSC 1 MC

Which of the following is equal to $(a + ib)^3$?

**A.** $(a^3 - 3ab^2) + i(3a^2b + b^3)$

**B.** $(a^3 + 3ab^2) + i(3a^2b + b^3)$

**C.** $(a^3 - 3ab^2) + i(3a^2b - b^3)$

**D.** $(a^3 + 3ab^2) + i(3a^2b - b^3)$

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5. Complex Numbers, EXT2 N1 2019 HSC 8 MC

Let $z$ be a complex number such that $z^2 = -i\overline{z}$.

Which of the following is a possible value for $z$?

**A.** $\dfrac{1}{2} - \dfrac{\sqrt{3}}{2}i$

**B.** $\dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i$

**C.** $\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}i$

**D.** $\dfrac{\sqrt{3}}{2} + \dfrac{1}{2}i$

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6. Complex Numbers, EXT2 N1 2020 SPEC2 8 MC

Given that $(x + iy)^{14} = a + ib$, where $x, y, a, b \in R$, $(y - ix)^{14}$ for all values of $x$ and $y$ is equal to

**A.** $-a - ib$

**B.** $-b + ia$

**C.** $-a + ib$

**D.** $b + ia$

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7. Complex Numbers, EXT2 N2 2020 SPEC2 5 MC

Given the complex number $z = a + bi$, where $a \in R\backslash\{0\}$ and $b \in R$, $\dfrac{4z\overline{z}}{(z + \overline{z})^2}$ is equivalent to

**A.** $1 + \left(\dfrac{\mathrm{Im}(z)}{\mathrm{Re}(z)}\right)^2$

**B.** $4[\mathrm{Re}(z) \times \mathrm{Im}(z)]$

**C.** $4\left([\mathrm{Re}(z)]^2 + [\mathrm{Im}(z)]^2\right)$

**D.** $\dfrac{2 \times \mathrm{Im}(z)}{[\mathrm{Re}(z)]^2}$

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## 8. Complex Numbers, EXT2 N1 2015 HSC 11a

Express $\dfrac{4+3i}{2-i}$ in the form $x+iy$, where $x$ and $y$ are real. ***(2 marks)***

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## 9. Complex Numbers, EXT2 N1 2011 HSC 2a

Let $w=2-3i$ and $z=3+4i$.

i. Find $\overline{w}+z$. ***(1 mark)***

ii. Find $|\,w\,|$. ***(1 mark)***

iii. Express $\dfrac{w}{z}$ in the form $a+ib$, where $a$ and $b$ are real numbers. ***(2 marks)***

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## 10. Complex Numbers, EXT2 N1 2018 HSC 11a

Let $z=2+3i$ and $w=1-i$.

i. Find $zw$. ***(1 mark)***

ii. Express $\overline{z}-\dfrac{2}{w}$ in the form $x+iy$, where $x$ and $y$ are real numbers. ***(2 marks)***

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## 11. Complex Numbers, EXT2 N1 2019 HSC 11a

Let $z=1+3i$ and $w=2-i$.

i. Find $z+\overline{w}$. ***(1 mark)***

ii. Express $\dfrac{z}{w}$ in the form $x+iy$, where $x$ and $y$ are real numbers. ***(2 marks)***

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## 12. Complex Numbers, EXT2 N1 2007 HSC 2a

Let $z=4+i$ and $w=\overline{z}$. Find, in the form $x+iy$,

i. $w$ ***(1 mark)***

ii. $w-z$ ***(1 mark)***

iii. $\dfrac{z}{w}$. ***(1 mark)***

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## 13. Complex Numbers, EXT2 N1 2008 HSC 2a

Find real numbers $a$ and $b$ such that $(1+2i)(1-3i)=a+ib$. ***(2 marks)***

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## 14. Complex Numbers, EXT2 N1 2009 HSC 2a

Write $i^9$ in the form $a+ib$ where $a$ and $b$ are real. ***(1 mark)***

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## 15. Complex Numbers, EXT2 N1 2009 HSC 2b

Write $\dfrac{-2+3i}{2+i}$ in the form $a+ib$ where $a$ and $b$ are real. ***(1 mark)***

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## 16. Complex Numbers, EXT2 N1 2010 HSC 2a

Let $z=5-i$.

i. Find $z^2$ in the form $x+iy$. ***(1 mark)***

ii. Find $z+2\overline{z}$ in the form $x+iy$. ***(1 mark)***

iii. Find $\dfrac{i}{z}$ in the form $x+iy$. ***(2 marks)***

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## 17. Complex Numbers, EXT2 N1 2012 HSC 11a

Express $\dfrac{2\sqrt{5}+i}{\sqrt{5}-i}$ in the form $x+iy$, where $x$ and $y$ are real. ***(2 marks)***

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## 18. Complex Numbers, EXT2 N1 2021 HSC 11b

Find $\displaystyle\sum_{n=1}^{5}(i)^n$. ***(2 marks)***

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## 19. Complex Numbers, EXT2 N1 2005 HSC 2a

Let $z=3+i$ and $w=1-i$. Find, in the form $x+iy$,

i. $2z+iw$ ***(1 mark)***

ii. $\overline{z}w$ ***(1 mark)***

iii. $\dfrac{6}{w}$ ***(1 mark)***

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## 20. Complex Numbers, EXT2 N1 2020 HSC 11a

Consider the complex numbers $w=-1+4i$ and $z=2-i$.

i. Evaluate $|w|$. ***(1 mark)***

ii. Evaluate $w\overline{z}$. ***(2 marks)***

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## 21. Complex Numbers, EXT2 N1 2022 HSC 11a

Express $\dfrac{3-i}{2+i}$ in the form $x+iy$, where $x$ and $y$ are real numbers. ***(2 marks)***

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## 22. Complex Numbers, EXT1 N1 SM-Bank 5

Let $z = 1 + 2i$ and $w = 3 - i$.

Find, in the form $x + iy$,

i. $zw$ *(1 mark)*

ii. $\overline{\left(\dfrac{10}{z}\right)}$. *(1 mark)*

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## 23. Complex Numbers, EXT2 N1 2006 HSC 2a

Let $z = 3 + i$ and $w = 2 - 5i$. Find, in the form $x + iy$,

i. $z^2$. *(1 mark)*

ii. $\overline{z}w$. *(1 mark)*

iii. $\dfrac{w}{z}$. *(1 mark)*

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## 24. Complex Numbers, EXT2 N1 2021 HSC 11e

The complex numbers $z = 5 + i$ and $w = 2 - 4i$ are given.

Find $\dfrac{\overline{z}}{w}$, giving your answer in Cartesian form. *(2 marks)*

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## 25. Complex Numbers, EXT2 N1 EQ-Bank 2

Find the values of $z$, in the form $z = a + ib$, such that

$z = \sqrt{7 + 24i}$ *(2 marks)*

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## 26. Complex Numbers, EXT2 N1 EQ-Bank 3

Find the values of $z$, in the form $z = x + iy$, such that

$z = \sqrt{-15 + 8i}$ *(2 marks)*

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## 27. Complex Numbers, EXT2 N2 2021 HSC 11d

i. Find the two square roots of $-i$, giving the answers in the form $x + iy$, where $x$ and $y$ are real numbers. *(2 marks)*

ii. Hence, or otherwise, solve $z^2 + 2z + 1 + i = 0$ giving your solutions in the form $a + ib$ where $a$ and $b$ are real numbers. *(2 marks)*

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# Worked Solutions

## 1. Complex Numbers, EXT2 N1 2012 HSC 1 MC

$$\begin{aligned} 2z + \overline{w} &= 2(5 - i) + 2 - 3i \\ &= 10 - 2i + 2 - 3i \\ &= 12 - 5i \end{aligned}$$

$\Rightarrow D$

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## 2. Complex Numbers, EXT2 N1 2015 HSC 2 MC

$$\begin{aligned} (4–3i)^2 &= 16–24i + 9i^2 \\ &= 7–24i \end{aligned}$$

$\Rightarrow\ A$

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## 3. Complex Numbers, EXT2 N1 2019 HSC 1 MC

$$\begin{aligned} (3–2i)^2 &= 9–12i + 4i^2 \\ &= 9–12i–4 \\ &= 5–12i \end{aligned}$$

$\Rightarrow\ A$

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## 4. Complex Numbers, EXT2 N1 2023 HSC 1 MC

$$\begin{aligned} (a + ib)^3 &= a^3 + 3a^2ib + 3a(ib)^2 + (ib)^3 \\ &= a^3 + 3a^2ib - 3ab^2 - ib^3 \\ &= (a^3 - 3ab^2) + i(3a^2b - b^3) \end{aligned}$$

$\Rightarrow C$

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5. Complex Numbers, EXT2 N1 2019 HSC 8 MC

**Solution 1**
Consider $C$:

$$z = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$
$$\overline{z} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$
$$-i\overline{z} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$\begin{aligned} z^2 &= \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)^2 \\ &= \frac{3}{4} - 2\frac{\sqrt{3}}{2} \cdot \frac{1}{2}i - \frac{1}{4} \\ &= \frac{1}{2} - \frac{\sqrt{3}}{2}i \end{aligned}$$

$\Rightarrow C$

**Solution 2**
Let $\arg(z) = \theta$

$$\arg\left(z^2\right) = 2\theta$$

$$\begin{aligned} \arg(-i\overline{z}) &= \arg(i\overline{z}) - \pi \\ &= \arg(\overline{z}) - \pi + \frac{\pi}{2} \\ &= -\theta - \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} \therefore 2\theta &= -\theta - \frac{\pi}{2} \\ 3\theta &= -\frac{\pi}{2} \\ \theta &= -\frac{\pi}{6} \end{aligned}$$

$\Rightarrow \ C$

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6. Complex Numbers, EXT2 N1 2020 SPEC2 8 MC

$$\begin{aligned} (y - ix)^{14} &= (-i(x + iy))^{14} \\ &= -i^{14}(x + iy)^{14} \\ &= -(x + iy)^{14} \\ &= -a - ib \end{aligned}$$

$\Rightarrow A$

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7. Complex Numbers, EXT2 N2 2020 SPEC2 5 MC

$z = a + ib, \ \overline{z} = a - ib$

$$\begin{aligned} \frac{4z\overline{z}}{(z + \overline{z})^2} &= \frac{4\left(a^2 + b^2\right)}{4a^2} \\ &= 1 + \left(\frac{b}{a}\right)^2 \\ &= 1 + \left(\frac{\text{Im}(z)}{\text{Re}(z)}\right)^2 \end{aligned}$$

$\Rightarrow A$

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8. Complex Numbers, EXT2 N1 2015 HSC 11a

$$\begin{aligned} \frac{4 + 3i}{2 - i} &= \frac{4 + 3i}{2 - i} \times \frac{2 + i}{2 + i} \\ &= \frac{8 + 4i + 6i - 3}{4 + 1} \\ &= \frac{5 + 10i}{5} \\ &= 1 + 2i \end{aligned}$$

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9. Complex Numbers, EXT2 N1 2011 HSC 2a

**i.** $w = 2 – 3i, \quad z = 3 + 4i, \quad \overline{w} = 2 + 3i$

$$\begin{aligned} \overline{w} + z &= 2 + 3i + 3 + 4i \\ &= 5 + 7i \end{aligned}$$

**ii.** $$\begin{aligned} |\,w\,| &= \sqrt{2^2 + 3^2} \\ &= \sqrt{13} \end{aligned}$$

**iii.** $$\begin{aligned} \frac{w}{z} &= \frac{2–3i}{3+4i} \times \frac{3–4i}{3–4i} \\ &= \frac{6–8i–9i–12}{9+16} \\ &= \frac{-6–17i}{25} \\ &= -\frac{6}{25} - \frac{17}{25}i \end{aligned}$$

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10. Complex Numbers, EXT2 N1 2018 HSC 11a

**i.** $z = 2 + 3i \quad w = 1 – i$

$$\begin{aligned} zw &= (2+3i)(1–i) \\ &= 2–2i + 3i–3i^2 \\ &= 2 + i + 3 \\ &= 5 + i \end{aligned}$$

♦ *Mean mark part (i) 97!%.*

**ii.** $$\begin{aligned} \overline{z} – \frac{2}{w} &= 2–3i – \frac{2}{1–i} \\ &= 2–3i – \frac{2(1–i)}{(1–i)(1+i)} \\ &= 2–3i–(1+i) \\ &= 1–4i \end{aligned}$$

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11. Complex Numbers, EXT2 N1 2019 HSC 11a

**i.** $z = 1 + 3i$

$$\begin{aligned} w &= 2–i \;\Rightarrow\; \overline{w} = 2 + i \\ z + \overline{w} &= 1 + 3i + 2 + i \\ &= 3 + 4i \end{aligned}$$

**ii.** $$\begin{aligned} \frac{z}{w} &= \frac{1+3i}{2–i} \times \frac{2+i}{2+i} \\ &= \frac{(1+3i)(2+i)}{2^2–i^2} \\ &= \frac{2 + i + 6i + 3i^2}{5} \\ &= \frac{-1+7i}{5} \\ &= -\frac{1}{5} + \frac{7}{5}i \end{aligned}$$

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12. Complex Numbers, EXT2 N1 2007 HSC 2a

**i.** $z = 4 + i, \quad w = \overline{z} = 4–i$

**ii.** $$\begin{aligned} w–z &= 4–i–(4+i) \\ &= -2i \end{aligned}$$

**iii.** $$\begin{aligned} \frac{z}{w} &= \frac{4+i}{4–i} \times \frac{4+i}{4+i} \\ &= \frac{(4+i)^2}{16+1} \\ &= \frac{16 + 8i - 1}{17} \\ &= \frac{15}{17} + \frac{8}{17}i \end{aligned}$$

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13. Complex Numbers, EXT2 N1 2008 HSC 2a

$$\begin{aligned} (1+2i)(1-3i) &= 1–3i + 2i–6i^2 \\ &= 1–i + 6 \\ &= 7–i \end{aligned}$$

$$\therefore a = 7, \; b = -1$$

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14. Complex Numbers, EXT2 N1 2009 HSC 2a

$$\begin{aligned} i^9 &= i \times i^8 \\ &= i \\ &= 0 + 1i \end{aligned}$$

15. Complex Numbers, EXT2 N1 2009 HSC 2b

$$\begin{aligned} &\frac{-2+3i}{2+i} \times \frac{2-i}{2-i} \\ &= \frac{(-2+3i)(2–i)}{4+1} \\ &= \frac{-4+2i+6i+3}{5} \\ &= \frac{8i–1}{5} \\ &= -\frac{1}{5} + \frac{8}{5}i \end{aligned}$$

16. Complex Numbers, EXT2 N1 2010 HSC 2a

**i.** $z = 5 - i$

$$\begin{aligned} \therefore z^2 &= (5-i)^2 \\ &= 25 - 10i + i^{\,2} \\ &= 24 - 10i \end{aligned}$$

**ii.** $z + 2\overline{z} = 5 - i + 2(5+i)$

$$= 15 + i$$

**iii.**
$$\begin{aligned} \frac{i}{z} &= \frac{i}{5-i} \times \frac{5+i}{5+i} \\ &= \frac{i(5+i)}{25+1} \\ &= \frac{5i-1}{26} \\ &= -\frac{1}{26} + \frac{5}{26}i \end{aligned}$$

17. Complex Numbers, EXT2 N1 2012 HSC 11a

$$\begin{aligned} \frac{2\sqrt{5}+i}{\sqrt{5}-i} &= \frac{\left(2\sqrt{5}+i\right)\left(\sqrt{5}+i\right)}{\left(\sqrt{5}-i\right)\left(\sqrt{5}+i\right)} \\ &= \frac{10+2\sqrt{5}i+\sqrt{5}i-1}{5+1} \\ &= \frac{9+3\sqrt{5}i}{6} \\ &= \frac{3}{2} + \frac{\sqrt{5}}{2}i \end{aligned}$$

18. Complex Numbers, EXT2 N1 2021 HSC 11b

$$\begin{aligned} \sum_{n=1}^{5} (i)^n &= i + i^2 + i^3 + i^4 + i^5 \\ &= i–1–i + 1 + i \\ &= i \end{aligned}$$

19. Complex Numbers, EXT2 N1 2005 HSC 2a

**i.** $z = 3 + i\ ,\ w = 1–i$

$$\begin{aligned} 2z + iw &= 2(3+i) + i(1–i) \\ &= 6 + 2i + i–i^2 \\ &= 7 + 3i \end{aligned}$$

**ii.**
$$\begin{aligned} \overline{z}w &= (3–i)(1–i) \\ &= 3–3i–\,i + i^2 \\ &= 2–4i \end{aligned}$$

**iii.**
$$\begin{aligned} \frac{6}{w} &= \frac{6}{1–i} \times \frac{1+i}{1+i} \\ &= \frac{6+6i}{1^2–i^2} \\ &= \frac{6+6i}{2} \\ &= 3 + 3i \end{aligned}$$

20. Complex Numbers, EXT2 N1 2020 HSC 11a

**i.** $w = -1 + 4i$

$|w| = \sqrt{(-1)^2 + 4^2} = 17$

**ii.** $z = 2 - i$

$\overline{z} = 2 + i$

$w\overline{z} = (-1 + 4i)(2 + i)$

$= -2 - i + 8i + 4i^2$

$= -6 + 7i$

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21. Complex Numbers, EXT2 N1 2022 HSC 11a

$$\frac{3-i}{2+i} = \frac{3-i}{2+i} \times \frac{2-i}{2-i}$$

$$= \frac{6 - 3i - 2i + i^2}{2^2 - i^2}$$

$$= \frac{5 - 5i}{5}$$

$$= 1 - i$$

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22. Complex Numbers, EXT1 N1 SM-Bank 5

**i.** $zw = (1 + 2i)(3 - i)$

$= 3 - i + 6i - 2i^2$

$= 5 + 5i$

**ii.** $$\frac{10}{z} = \frac{10}{1+2i} \times \frac{1-2i}{1-2i}$$

$$= \frac{10 - 20i}{1^2 - (2i)^2}$$

$$= \frac{10 - 20i}{1 + 4}$$

$$= 2 - 4i$$

$$\therefore \overline{\frac{10}{z}} = 2 + 4i$$

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23. Complex Numbers, EXT2 N1 2006 HSC 2a

**i.** $z^2 = (3 + i)^2$

$= 9 + 6i - 1$

$= 8 + 6i$

**ii.** $\overline{z}w = (3 - i)(2 - 5i)$

$= 6 - 15i - 2i - 5$

$= 1 - 17i$

**iii.** $$\frac{w}{z} = \frac{2-5i}{3+i} \times \frac{3-i}{3-i}$$

$$= \frac{1-17i}{10}$$

$$= \frac{1}{10} - \frac{17}{10}i$$

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24. Complex Numbers, EXT2 N1 2021 HSC 11e

$z = 5 + i \ \Rightarrow \ \overline{z} = 5 - i$

$w = 2 - 4i$

$$\frac{\overline{z}}{w} = \frac{5-i}{2-4i} \times \frac{2+4i}{2+4i}$$

$$= \frac{(5-i)(2+4i)}{4+16}$$

$$= \frac{10 + 20i - 2i + 4}{20}$$

$$= \frac{7}{10} + \frac{9}{10}i$$

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## 25. Complex Numbers, EXT2 N1 EQ-Bank 2

$z = \sqrt{7 + 24i}$
$z^2 = 7 + 24i$
$7 + 24i = (a + ib)^2$
$= a^2 - b^2 + 2abi$

$a^2 - b^2 = 7 \ldots (1)$
$2ab = 24$
$ab = 12 \ldots (2)$

$a = 4, \quad b = 3$
$a = -4, \quad b = -3$

$\therefore z_1 = 4 + 3i$
$z_2 = -4 - 3i$

## 26. Complex Numbers, EXT2 N1 EQ-Bank 3

$z = \sqrt{-15 + 8i}$
$z^2 = -15 + 8i$
$-15 + 8i = (x + iy)^2$
$= x^2 - y^2 + 2xyi$

$x^2 - y^2 = -15 \ldots (1)$
$2xy = 8$
$xy = 4 \ldots (2)$

$x = 1, \quad y = 4$
$x = -1, \quad y = -4$

$\therefore z_1 = 1 + 4i$
$z_2 = -1 - 4i$

## 27. Complex Numbers, EXT2 N2 2021 HSC 11d

**i.** Solution 1

$z = x + iy\ ,\ z^2 = -i$
$z^2 = x^2 - y^2 + 2xyi = -i$
$x^2 - y^2 = 0 \ldots (1)$
$2xy = -1 \ldots (2)$
$x = \pm y \ldots (1)'$

Substitute $x = y$ into $(2)$
$2x^2 = -1 \rightarrow$ no real solutions
Substitute $x = -y$ into $(2)$

$-2x^2 = -1$

$x = \pm \dfrac{1}{\sqrt{2}}$

$\therefore\ z = \dfrac{1}{\sqrt{2}} - \dfrac{1}{\sqrt{2}}i$ or $z = -\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}}i$

Solution 2
$z = re^{i\theta}\ ,\ z^2 = -i$
$r^2 e^{i2\theta} = e^{i\frac{3\pi}{2}}$ or $e^{-i\frac{\pi}{2}}$
$\Rightarrow\ r = 1\ ,\ \theta = \dfrac{3\pi}{4}$ or $-\dfrac{\pi}{4}$

$z = \cos\dfrac{3\pi}{4} + i\sin\dfrac{3\pi}{4} = -\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}}i$
$z = \cos\left(-\dfrac{\pi}{4}\right) + i\sin\left(-\dfrac{\pi}{4}\right) = \dfrac{1}{\sqrt{2}} - \dfrac{1}{\sqrt{2}}i$

**ii.** $z^2 + 2z + 1 + i = 0$

$z = \dfrac{2 \pm \sqrt{4 - 4 \cdot 1 \cdot (1 + i)}}{2}$
$= \dfrac{-2 \pm \sqrt{4 - 4 - 4i}}{2}$
$= -1 \pm \sqrt{-i}$

$\therefore\ z = -1 + \dfrac{1}{\sqrt{2}} - \dfrac{1}{\sqrt{2}}i$ or $z = -1 - \dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}}i$